

A Guide to Statistical Network Modeling

Session 2.2: Separable TERGMs for Formation and Dissolution

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instats Seminar

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Companion computations and reproducibility

- **Shared resources:** [repository](#) and [companion app](#); see Session 1.1 for the common reproducibility workflow.
- Open **Session 2.2** and choose one of **five visible public temporal-network workflows**; each exposes formation and persistence supports, component estimates, all audit plots, and a downloadable record.
- **Computation boundary:** the app fits a transparent, dyad-independent baseline STERGM with separate intercept-only formation and persistence components; it does not silently claim a general MCMC-fitted STERGM.
- **This session:** separability, process-specific interpretation, duration, estimation, dynamic goodness of fit, five public examples, and a support-aware BYOD workflow.

Learning objectives

By the end of this session, we will be able to:

- define the **formation network** and **persistence network** for an observed transition;
- state the additional conditional-independence restriction that makes a TERGM **separable**;
- distinguish a persistence effect from a dissolution effect and identify their sign reversal;
- explain why cross-sectional prevalence can reflect formation, duration, or both; and
- audit formation–persistence components through support checks, rates, simulations, duration summaries, and appropriately limited uncertainty checks.

Session route

- ➊ **Objects and supports:** formation, persistence, dissolution, and endpoint reconstruction.
- ➋ **Separable models:** component distributions, process-specific effects, and observation intervals.
- ➌ **Duration and estimation:** geometric special case, censoring, conditional likelihood, and equilibrium alternatives.
- ➍ **Diagnostics:** dynamic goodness of fit for both states and transitions.
- ➎ **No-code practice:** five public repeated networks and a constrained separable BYOD calculation.

Why separate formation from persistence?

- A tie observed at time t can be prevalent because it **forms often**, because it **lasts long**, or because both are true.
- A combined transition coefficient can obscure whether a predictor changes **incidence** among prior non-ties or **survival** among prior ties.
- A static cross-section cannot identify the two mechanisms from prevalence alone.
- **Separable temporal ERGMs (STERGMs)** assign distinct stochastic components to the two risk sets.
- **Key question:** what creates ties that are absent, and what retains ties that already exist?

Observed transition notation

- Let Y^{t-1} and Y^t be two observed binary networks on their documented transition support $D_{t-1,t}$.
- Use the Session 2.1 four-cell table $(N_{00}, N_{01}, N_{10}, N_{11})$ on the documented joint risk set; here it is re-partitioned into formation and persistence supports.
- As before, overall stability $(N_{00} + N_{11})/|D_{t-1,t}|$ is not tie persistence $N_{11}/(N_{10} + N_{11})$.

Formation and persistence networks

- Define the **formation network** by $Y^+ = Y^{t-1} \cup Y^t$.
- Define the **persistence network** by $Y^- = Y^{t-1} \cap Y^t$.
- Y^+ contains every prior tie plus every newly formed tie; only prior non-ties can change within this component.
- Y^- contains only prior ties that survive; only prior ties can change within this component.
- The **dissolution set** is $Y^{\text{diss}} = Y^{t-1} \setminus Y^-$, not the same object as Y^- .

Endpoint reconstruction is deterministic

- Conditional on Y^{t-1} , the pair (Y^+, Y^-) determines the endpoint network.
- Specifically,

$$Y^t = Y^- \cup (Y^+ \setminus Y^{t-1}).$$

- Thus $Y^t \leftrightarrow (Y^+, Y^-)$ is a **bijection** between compatible endpoints and compatible formation–persistence pairs.
- The bijection is deterministic; it does **not** itself imply stochastic independence between the two components.
- A STERGMM adds a probabilistic factorization beyond this bookkeeping identity.

Formation support

- The formation component is supported on supergraphs of the prior network:

$$\mathcal{Y}^+(y^{t-1}) = \left\{ y^+ : y^+ \supseteq y^{t-1} \right\}.$$

- A dyad with $y_{ij}^{t-1} = 1$ is **not** a formation opportunity.
- Its states are fixed at one in Y^+ , so it contributes no formation toggle comparison.
- Actor absence, structural zeros, incidental missingness, and fixed design constraints must be removed before defining this support.
- **No recoding rule:** unavailable dyads are not zero ties and are not formation failures.

Persistence support

- The persistence component is supported on subgraphs of the prior network:

$$\mathcal{Y}^-(y^{t-1}) = \left\{ y^- : y^- \subseteq y^{t-1} \right\}.$$

- A dyad with $y_{ij}^{t-1} = 0$ is **not** at risk of persistence or dissolution.
- Its state is fixed at zero in Y^- , whereas a prior tie may survive or dissolve.
- The persistence risk set can be tiny in a sparse network even when the formation risk set is huge.
- Therefore equal numerical coefficients across the two components are neither required nor usually substantively expected.

The separability restriction

- A STERGM parameterizes formation and persistence separately, with variation-independent parameter spaces $(\theta^+, \theta^-) \in \Theta^+ \times \Theta^-$, and assumes that, conditional on Y^{t-1} ,

$$\mathbb{P}_{\theta^+, \theta^-}(Y^+, Y^- \mid Y^{t-1}) = \mathbb{P}_{\theta^+}(Y^+ \mid Y^{t-1})\mathbb{P}_{\theta^-}(Y^- \mid Y^{t-1}).$$

- The factorization requires compatible Cartesian-product support: every allowed pair must reconstruct to an allowed endpoint.
- Global constraints coupling formation and persistence can invalidate this factorization.
- **Separability is a model assumption**, not a consequence of observing a panel.

Separable does not mean dyad-independent

- Within a formation component, ties can still be dependent through closure, stars, homophily, or other network statistics.
- Within a persistence component, survival can likewise depend on embedding, degree, covariates, or prior structure.
- The independence assumption is **between components conditional on the previous network**, not necessarily among dyads within a component.
- General component dependence usually leaves an intractable normalizing constant and can require **MCMC-based estimation**.
- The app's baseline model is explicitly narrower: each component is dyad-factorizing and intercept-only.

Formation probability and change statistics

- A general formation component has conditional exponential-family form

$$\mathbb{P}_{\theta^+}(Y^+ = y^+ \mid Y^{t-1}) \propto \exp\{(\theta^+)^T g^+(y^+, Y^{t-1})\}.$$

- For an absent admissible dyad with both formation states feasible,

$$\log \frac{\mathbb{P}(Y_{ij}^+ = 1 \mid Y_{-ij}^+, Y^{t-1})}{\mathbb{P}(Y_{ij}^+ = 0 \mid Y_{-ij}^+, Y^{t-1})} = (\theta^+)^T \Delta_{ij}^+ g^+.$$

- A positive formation coefficient favors the associated **creation** configuration, conditional on the rest of the component model.

Persistence probability and change statistics

- A general persistence component is

$$\mathbb{P}_{\theta^-}(Y^- = y^- \mid Y^{t-1}) \propto \exp\{(\theta^-)^T g^-(y^-, Y^{t-1})\}.$$

- For an existing admissible tie with both persistence states feasible,

$$\log \frac{\mathbb{P}(Y_{ij}^- = 1 \mid Y_{-ij}^-, Y^{t-1})}{\mathbb{P}(Y_{ij}^- = 0 \mid Y_{-ij}^-, Y^{t-1})} = (\theta^-)^T \Delta_{ij}^- g^-.$$

- A positive persistence coefficient favors **retention**; its implication for dissolution is in the opposite direction.

Persistence is not the dissolution network

- Y^- records **surviving prior ties**; Y^{diss} records **prior ties that end**.
- A positive coefficient on a persistence statistic increases the chance of survival, hence decreases the chance of dissolution.
- In `tergm`, `Persist()` evaluates a statistic on Y^- .
- The `Diss()` operator negates a statistic evaluated on that same persistence network; it does not generally evaluate the set of dissolved ties as a separate graph.
- Interpret sign conventions from the operator and support, not from the informal word “dissolution.”

One predictor can have two process-specific effects

- A nodal similarity term can be positive for formation but near zero for persistence.
- Alternatively, a similarity relation may not create many new ties but may help existing ties endure.
- A closure term can create ties among actors with shared partners without prolonging the ties once formed.
- Conversely, embedded ties may survive longer even if new closure is weak.
- **Process-specific parameters** make these different hypotheses estimable in principle; cross-sectional clustering alone does not distinguish them.

Prevalence is a balance of incidence and duration

- High cross-sectional prevalence of a tie type may arise from high formation probability, high persistence probability, or both.
- For the dyad-independent edges-only model at equilibrium, let $f = \mathbb{P}(Y_{ij}^+ = 1 \mid Y_{ij}^{t-1} = 0)$ and $p = \mathbb{P}(Y_{ij}^- = 1 \mid Y_{ij}^{t-1} = 1)$. Then

$$p_{\text{cs}} = \frac{f}{f + 1 - p}, \quad \mathbb{E}(D) = \frac{1}{1 - p},$$

and hence

$$\theta_{\text{edges}}^+ = \text{logit} \left\{ \frac{p_{\text{cs}}}{\mathbb{E}(D)(1 - p_{\text{cs}})} \right\}.$$

- This homogeneous-equilibrium relationship does not generalize to arbitrary STERGMs.
- A cross-sectional association therefore cannot by itself identify a formation mechanism or a duration mechanism.

A simple support example

Prior Y_{ij}^{t-1}	Current Y_{ij}^t	Formation component	Persistence component
0	0	at risk, no formation	fixed zero
0	1	at risk, formation	fixed zero
1	0	fixed one	at risk, no persistence
1	1	fixed one	at risk, persistence

- The same endpoint transition is viewed through two different supports.
- **Formation events** are N_{01} ; **dissolution events** are N_{10} ; both must be counted on the joint observed risk set.
- The table makes clear why a single combined logistic regression is not the same model as separate components.

Observation intervals define the estimand

- As established in Session 2.1, the chosen interval defines the discrete-time estimand: endpoints can conceal within-interval events, and parameters do not automatically transfer across time scales.
- With a constant continuous-time dissolution hazard μ , uninterrupted survival over interval Δ is $\exp(-\mu\Delta)$; this need not equal $\mathbb{P}(Y_{ij}^t = 1 \mid Y_{ij}^{t-1} = 1)$ when unobserved within-interval reformation is possible.

Baseline duration model

- Suppose persistence is dyad-independent, time-homogeneous, and memoryless with per-period dissolution probability q .
- Let a tie that survives no additional observation interval have duration $D = 1$.
- Then

$$\mathbb{P}(D = d) = (1 - q)^{d-1}q, \quad d = 1, 2, \dots$$

- Equivalently, with persistence probability $p = 1 - q$, $\mathbb{P}(D \geq d) = p^{d-1}$ and $\mathbb{E}(D) = 1/q$.
- This **geometric duration** result is not automatic for a general STERGM.

Duration interpretation of a persistence intercept

- For an edges-only baseline persistence component,

$$\text{logit}(p) = \theta_{\text{edges}}^-.$$

- Hence

$$\mathbb{E}(D) = \frac{1}{1-p} = 1 + \exp\{\theta_{\text{edges}}^-\}.$$

- The app reports this expected duration only as a **panel-interval quantity** under the homogeneous memoryless baseline.
- Age dependence, time-varying covariates, structural persistence effects, or interval heterogeneity destroy this simple geometric interpretation.

Censoring is part of the duration audit

- A tie already present at the first observed wave has an **unknown prior age** (left-censored/left-truncated observation of its spell): its true formation time is not observed.
- A tie still present at the final observed wave is **right-censored**: its eventual dissolution time is unknown.
- A tie whose dyad leaves the observed support is **support-censored**; it is not known to dissolve.
- Empirical spell summaries must not treat these observations as complete lifetimes.
- The app plots only complete observed spells and reports the three censoring counts separately.

From general STERGMs to the implemented baseline

- A general STERGM may use endogenous statistics within g^+ and g^- , such as closure, degree, mixing, or covariate effects.
- The app implements instead

$$\text{logit } \mathbb{P}(Y_{ij}^+ = 1 \mid Y_{ij}^{t-1} = 0) = \theta_{\text{edges}}^+, \quad \text{logit } \mathbb{P}(Y_{ij}^- = 1 \mid Y_{ij}^{t-1} = 1) = \theta_{\text{edges}}^-.$$

- Each component is therefore a transparent exact conditional likelihood on its own risk set.
- The baseline is pedagogical and auditable; it is not mislabeled as a general network-dependent STERGM.

Conditional maximum likelihood estimation

- With observed successive networks, **conditional maximum likelihood estimation** fits the formation and persistence distributions conditional on the preceding observed networks, each on its constrained support.
- In a general component ERGM, the normalizing constant can be intractable; CMLE can require MCMC approximation.
- Separate component likelihoods do not turn a dependent STERGM into an ordinary independent-dyad regression.
- The amount of temporal replication still limits what can be learned about parameter homogeneity and uncertainty.
- The appropriate estimation method follows from the stochastic model and data design, not from software convenience.

Equilibrium generalized method of moments

- **EGMME** can be used when a cross-sectional network is accompanied by temporal information such as tie durations.
- It assumes the observed network is a realization of a TERGM process at **stationarity**.
- It matches equilibrium network and duration features rather than maximizing an observed-transition likelihood.
- EGMME and CMLE answer different questions and rely on different information.
- A cross-section plus duration data is not the same observation design as a panel of successive network transitions.

Estimation language that must not be conflated

- Use Session 2.1's distinction: **MCMLE** is MCMC likelihood approximation for a specified dependent model, whereas **MPLE** (including a bootstrap MPLE) is a pseudolikelihood procedure.
- Here, the app's dyad-factorizing intercept-only components have exact conditional likelihoods rather than MPLE; general dependent components can require MCMC.
- Whole-transition bootstrap intervals are withheld when too few transitions are available.

Component identifiability checks

- Formation requires both events and non-events among prior non-ties at risk.
- Persistence requires both survival and dissolution among prior ties at risk.
- A component with no outcome variation has no finite intercept-only logit coefficient.
- A large dyadic risk set does not replace having multiple observed transitions when the goal is to assess heterogeneity across intervals.
- The app displays support sizes, observed events, coefficient standard errors, numerical warnings, and the number of usable transitions before interpretation.

Dynamic goodness of fit: two targets

- A dynamic audit must assess **network states** and **transitions**.
- State summaries include tie totals, density, mean degree, degree distributions, shared partners, mixing, or triads where the fitted model makes these meaningful.
- Transition summaries include N_{01} formations, N_{10} dissolutions, N_{11} persistence, and $N_{00} + N_{11}$ stability.
- Tie-age or duration summaries are useful only with explicit censoring treatment.
- Matching a state-based statistic alone does not show that turnover is reproduced correctly.

One-step versus multi-step simulation

- **One-step simulation** conditions on each observed preceding network and asks whether the next observed state and transition summaries are plausible.
- **Multi-step simulation** propagates the fitted process and can reveal accumulated errors in density, turnover, mixing, or duration.
- A model may reproduce isolated endpoint states but still generate implausible longer-run trajectories.
- The app's baseline audit reports one-step conditional simulation envelopes; it does not overstate them as a complete multi-step process validation.
- Simulation-based fit is different from causal validation or a held-out predictive study.

All plots in the Session 2.2 app

- **Component support plot:** prior non-ties versus prior ties at every transition.
- **Coefficient plot:** formation and persistence intercepts with Wald intervals.
- **Rate plots:** observed formation and persistence rates versus homogeneous fitted probabilities.
- **One-step simulation plots:** ties, density, formations, dissolutions, persistence, stability, and mean degree.
- **Structural GOF plots:** degree/geodesics; ESP/DSP (undirected); in-/out-degree, reciprocity, and triads (directed); mixing if a categorical attribute is declared.
- **Duration plot:** complete observed spells with left-, right-, and support-censoring counts.
- **Bootstrap plot:** whole-transition sensitivity intervals only when sufficient transitions exist.
- For general network-dependent STERGMs, also audit component-specific MCMC trace, autocorrelation, and sampled-statistic distributions.

Worked examples: a deliberately limited public-data pool

- Each public example is a repeated **binary** network with a documented response meaning and a defensible transition support.
- The same public panels may be revisited across temporal sessions when the session-specific model is valid.
- The five examples span directed and undirected relations, multiple observation schedules, stable and changing actor sets, and different temporal limitations.
- We do **not** force ranked, fixed-choice, valued, cumulative, multiplex, or unknown-interval data into this baseline STERGM calculation.
- Public provenance, citations, source documentation, and local tables are visible in the app.

Worked example 1: Knecht classroom friendship

- **Data:** four directed binary classroom friendship panels from the public RSiena tutorial archive.
- **Support:** documented missing nominations, nonmembership, and departure are excluded rather than treated as zero ties or dissolution events.
- **Formation component:** good-friend nominations absent at the preceding observed school wave.
- **Persistence component:** previously reported good-friend nominations that remain observable at the next wave.
- **Limitation:** three transitions describe observed-wave change; they do not establish peer influence, microstep timing, or a general structural mechanism.

Worked example 2: Sampson monastery liking

- **Data:** three directed binary positive-affect/liking nomination panels among 18 monk-novitiates, distributed by Statnet.
- **Support:** the same labelled actors and the same relation layer are retained across the documented panels.
- **Use:** compare the baseline probability of creating a previously absent liking nomination with survival of an existing nomination.
- **Audit:** two one-step transitions, process-specific supports, rates, and censored spell summaries.
- **Limitation:** two transitions cannot identify temporal heterogeneity or support a causal behavioral response claim.

Worked example 3: Coleman high-school nominations

- **Data:** public Fall and Spring directed nomination panels among 73 high-school boys.
- **Support:** the fixed roster is a stated source-data assumption because no wave-specific attendance file is supplied.
- **Use:** one Fall-to-Spring transition illustrates that formation and persistence supports remain distinct even with minimal temporal replication.
- **Audit:** separate support counts, formation and survival rates, and one conditional simulation comparison.
- **Limitation:** a single transition cannot furnish bootstrap uncertainty, temporal trend, or observed duration inference.

Worked example 4: COW formal alliances

- **Data:** eleven annual undirected alliance panels, 1960–1970, from Correlates of War formal alliances and state-system membership data.
- **Response:** binary treaty presence for an unordered state pair; not treaty count, defence-only status, or a continuous cooperation score.
- **Support:** state entry and exit create structural unavailability and are excluded from transition risk sets.
- **Use:** ten annual transitions permit support, rate, one-step, duration, and whole-transition bootstrap audits.
- **Limitation:** annual endpoints do not reveal within-year ordering, dissolution-reformation sequences, or geopolitical causes.

Worked example 5: Windsurfers interaction

- **Data:** public repeated undirected binary beach-day interaction panels distributed by Statnet `networkDynamic`.
- **Support:** only pairs jointly present on adjacent observed days are at risk; beach absence is not an observed non-tie.
- **Gap:** the documented missing day is not bridged as one long transition.
- **Use:** daily formation and persistence supports, one-step simulations, mean-degree checks, and support-censored spell accounting.
- **Limitation:** observed day-level interaction is not necessarily a durable friendship state or a homogeneous duration process.

Worked-example navigation in the app

- 1 Open the **Five worked public networks** tab and select one of the five visible examples.
- 2 Read its response boundary, support note, method appropriateness, and interpretation limit before fitting.
- 3 Inspect the wave table and four-cell transition table; verify what counts as N_{01} , N_{10} , and N_{11} .
- 4 Open the reproducible settings only to choose a seed, number of conditional simulations, and, where justified, bootstrap resamples.
- 5 Fit the stated baseline model and inspect every component-support, rate, simulation, duration, and bootstrap panel in order.

BYOD: required data contract

- Use the Session 2.1 repeated-network input contract: `wave`, `id` nodes; `wave`, `source`, `target` loopless binary ties; optional `transition_block`; and a wave-level risk table whenever presence does not establish observability.
- Session 2.2 then validates directionality, duplicates, endpoints, minimum waves, and formation–persistence risk-set compatibility.
- Valued/count ties, rankings, multiplex layers, unknown time spacing, and actor absence coded as a non-tie are rejected rather than silently transformed.

BYOD: interpretation boundary

- The BYOD computation fits a **baseline dyad-factorizing STERGM**: formation intercept on prior non-ties and persistence intercept on prior ties.
- It produces exact component likelihoods only because the stated baseline has no endogenous within-component statistics.
- It does **not** identify causal effects, continuous-time hazards, unobserved microsteps, or general structural dependence.
- A positive persistence estimate means survival is more likely under the baseline; it does not independently estimate a separate dissolution-network coefficient.
- A completed calculation is a reproducible conditional model statement, not a certificate that the model is scientifically adequate.

TERGM versus STERGM: a model-choice guide

Question	Combined transition TERGM	Separable TERGM
One transition mechanism?	Models $Y^t \mid Y^{t-1}$ directly.	Decomposes formation and persistence.
Prior non-ties and ties?	May share coefficients and summaries.	Receive separate process supports.
Interpretation	Temporal association conditional on history.	Process-specific incidence and retention association.
Extra assumption	No separability required.	Conditional component factorization required.

- Choose the model whose stochastic object, support, and process interpretation match the research question and observation design.

Take-home synthesis

- **Separate supports first:** formation concerns prior non-ties; persistence concerns prior ties.
- **Keep objects distinct:** persistence network, dissolution set, stability, duration, and cross-sectional prevalence are not interchangeable.
- **Honor the interval:** observed endpoints do not reveal every within-interval event.
- **Audit processes and states:** formation, dissolution, persistence, stability, current network summaries, duration censoring, and simulation envelopes each answer different questions.
- **State the boundary:** a transparent baseline component model is valuable precisely when it is not overstated as a general network-dependent STERGM.

Selected sources

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